

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

STATION:	SALEM		
SYSTEM:	Reactor Coolant System		
TASK:	Measure Leakage to the Containment Sump		
TASK NUMBER:	N0020080101		
JPM NUMBER:	13-01 NRC RO A1-1		
ALTERNATE PATH:	☐	K/A NUMBER:	2.1.25
APPLICABILITY:	IMPORTANCE FACTOR:		
EO ☐	RO ☒	STA ☐	SRO ☐
		3.9	4.2
		RO	SRO
EVALUATION SETTING/METHOD:	Classroom		
REFERENCES:	S2.OP-AR.ZZ-0003, Overhead Annunciators Window C, Rev. 17 S2.OP-SO.RC-0004, Identifying and Measuring Leakage, Rev. 15 (Both Rev checked 9-26-14)		
TOOLS AND EQUIPMENT:	Calculator		
VALIDATED JPM COMPLETION TIME:	15 min		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	N/A		
Developed By:	G Gauding Instructor	Date:	9-26-14
Validated By:	D Tait SME or Instructor	Date:	10-15-14
Approved By:	 Training Department	Date:	11-18-14
Approved By:	 Operations Representative	Date:	11-18-14
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE:	☐ SAT	☐ UNSAT	
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:		DATE:	

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DATE: _____

SYSTEM: Reactor Coolant System

TASK: Measure Leakage to the Containment Sump

TASK NUMBER: N0020080101

SIMULATOR SETUP N/A

INITIAL CONDITIONS:

Unit 2 is operating at 100% power, steady state, BOL.

At 0750, OHA C-2 CNTMT SUMP PMP START unexpectedly annunciates, along with OHA A-41. The Aux typewriter shows 21 containment sump pump has started. 2 minutes later, OHA C-2 clears, OHA A-41 annunciates, and the Aux typewriter shows 21 containment sump pump has stopped. The plant remains stable, no indications of a RCS leak are apparent, and no operations have been performed which would cause containment sump level to rise.

INITIATING CUE:

You are the RO. Calculate the Containment Sump leak rate IAW S2.OP-SO.RC-0004, Identifying and Measuring Leakage, Section 5.3. Assume any required IV's will be performed SAT if needed.

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made (and NRC concurrence is obtained).

Task Standard for Successful Completion:

1. Operator calculates leakage into Containment Sump of 0.3 gpm.

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*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
*	SO.RC-4 5.3.1	Each time a Containment Sump Pump starts, RECORD the following on Attachment 1, Section 2.0: - Operating Containment Sump Pump Number - Time of pump start	Records "21 cont sump". Records time of pump start		
*	SO.RC-4 5.3.2	CALCULATE elapsed time in minutes between start time and previous Containment Sump Pump stop time AND RECORD on Attachment 1, Section 2.0.	Calculates elapsed time in minutes between start time and previous Containment Sump Pump stop time and records on Attachment 1, Section 2.0. by using previous stop time of <u>1510 yesterday</u> , and with cue of <u>0750</u> today determines elapsed time is 1,000 minutes .		
	SO.RC-4 5.3.3	ENSURE no draining, sampling, or liquid additions to Containment Sump have occurred during selected time frame.	Determines from initial conditions that no draining, sampling, or liquid additions to Containment Sump have occurred during selected time frame.		
*	SO.RC-4 5.3.4	CALCULATE Containment Sump Leak Rate using Attachment 3.	Calculates Containment Sump Leak Rate using Attachment 3 by using page 1 of 2 of Att. 3, and determines the 1,000 minute line crosses the 0.3 gpm leak rate line.		

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*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
	SO.RC-4 5.3.5	RECORD calculated Containment Sump leak rate on Attachment 1, Section 2.0.	Records calculated Containment Sump leak rate on Attachment 1, Section 2.0. as 0.3 gpm.		
	SO.RC-4 5.3.6	Direct a second Operator to PERFORM Independent Verification of the calculation(s) performed in Attachment 1, Section 2.0.	Determines from initiating cue that all IV's required to be performed will be performed SAT.		
	SO.RC-4 5.3.7	RECORD Containment Sump Pump stop time and date on Attachment 1, Section 2.0.	Records Containment Sump Pump stop time and date on Attachment 1, Section 2.0.		
	SO.RC-4 5.3.8	<u>IF</u> leakage to Containment Sump exceeds 1.0 gpm <u>AND</u> Unit in Modes 1-4, <u>THEN</u> : A. INITIATE S2.OP-ST.RC-0008(Q), Reactor Coolant Water Inventory Balance. B. REFER to Technical Specification 3.4.7.2.	Determines leakage to Containment Sump does not exceed 1.0 gpm		
	SO.RC-4 5.3.9	<u>IF</u> leakage to Containment Sump exceeds 0.85 gpm, <u>THEN</u> INITIATE Section 5.6 of this procedure.	Determines leakage to Containment Sump does not exceed 0.85 gpm.		
			Terminate JPM when procedure is returned to evaluator.		

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JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- 8 1. Task description and number, JPM description and number are identified.
- 8 2. Knowledge and Abilities (K/A) references are included.
- 8 3. Performance location specified. (in-plant, control room, or simulator)
- 8 4. Initial setup conditions are identified.
- 8 5. Initiating and terminating Cues are properly identified.
- 8 6. Task standards identified and verified by SME review.
- 8 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- 8 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 15 Date 10/15/14
- 8 9. Pilot test the JPM:
 - a. verify Cues both verbal and visual are free of conflict, and
 - b. ensure performance time is accurate.
- _____ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: 

Date: 10/15/14

SME/Instructor: _____

Date: _____

SME/Instructor: _____

Date: _____

**INITIAL
CONDITIONS:**

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